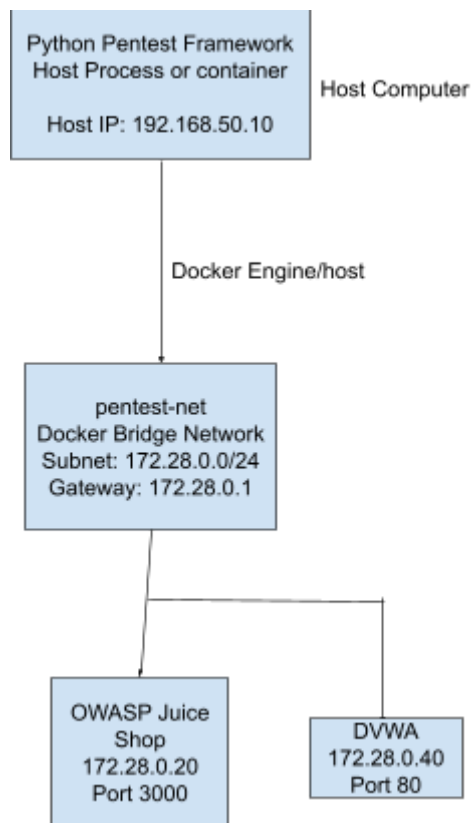


The network architecture I plan to implement will provide a controlled and secure Docker-based environment for developing and testing the framework. The environment will contain the Python penetration-testing framework and intentionally vulnerable web applications such as OWASP Juice Shop and DVWA.

Instead of using separate physical devices, the proposed implementation will use Docker containers connected through a dedicated user-defined Docker bridge network. This provides logical network isolation while allowing the containers to communicate with one another using private IP addresses and Docker DNS service names.



Docker environment will use the private IPv4 network

Docker Network: `pentest-net`

Network: 172.28.0.0/24

Subnet Mask: 255.255.255.0

Docker Gateway: 172.28.0.1

Broadcast address: 172.28.0.255

The /24 subnet is enough as the environment will only have a few containers.

[illegible]